

Design Criteria Report — Final (Base Case)

Salt + Water Suspension System — 10% (w/w) salt slurry

Throughput: **100 short tons/hr (200,000 lb/hr)** dry table salt → 10% mass slurry in a **10,000 US-gal** covered stainless tank, pumped downstream.

Location: Charlotte, NC (preliminary geotechnical assumption used).

Report status: Final base-case design criteria and preliminary cost & civil/utility addendum (includes updated civil, stormwater, grading, containment, survey, geotech, instrumentation-air, and safety system costs). All costs are order-of-magnitude planning estimates in 2025 USD unless noted. A 30% contingency is applied to the grand total.

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1. Executive summary

Design for a system to accept **100 short tons/hr (200,000 lb/hr)** of table salt delivered on an inclined troughed belt conveyor and produce a **10% (w/w)** salt-in-water slurry in a **10,000 US gallon** covered 304-SS tank elevated **5 ft** above final grade. Slurry pumped at design TDH = **100 ft**, base-case flow ≈ **3,733 GPM**. Prime mover selections: pump motor **175 HP VFD**, conveyor motor **15 HP fixed-speed**, agitator **15 HP fixed-speed**. Controls: local PLC + HMI with field instrumentation (level, temperature, flow, pressure, belt scale, safety interlocks). This final report incorporates a detailed civil & utilities addendum: earthwork cut/fill estimates, grading & pavement, stormwater conceptual design (1-inch capture basis), culvert and swale recommendations, secondary containment sized to 110% of tank volume, survey & geotechnical budgets, and an instrument-air compressor package. Updated cost tables reflect

these additions; grand total (with 30% contingency) $\approx$ **\$1,147,380** (includes optional fire pump allowance; remove optional fire pump to reduce cost by $\sim$ \$50k).

2. Design basis & explicit assumptions (all imperial units)

(As already developed; key items summarized and unchanged except for additions noted)

- Feed: **100 st/hr = 200,000 lb/hr** dry NaCl.
 - Particle size: **100–500 μm** . Salt moisture $\leq 0.5\%$.
 - NaCl true density: **135.6 lb/ft³**. Bulk density (loose): **$\sim$ 74.91 lb/ft³**.
 - Slurry concentration (design): **10% mass**. Process temperature 60 °F. 10% NaCl solution density **$\sim$ 8.93 lb/gal** (SG $\approx$ 1.07).
 - Conveyor: 24 in belt (2 ft), incline 10°, horizontal projection 100 ft, nominal belt speed 197 ft/min.
 - Tank: 10,000 US gal, 304 SS, covered, elevated on steel platform 5 ft above grade, H:D = 2:1.
 - Pump: centrifugal slurry pump, 316 SS wetted parts, pump efficiency assumed 70%, motor efficiency 95%, preliminary motor **175 HP VFD**. Suction piping recommended 24 in, discharge 20 in (confirm with vendor).
 - Geotech/Foundations initial allowable bearing = **3,000 psf** (Charlotte assumption). Final geotechnical required.
 - Existing site slope assumed **0.25%** (0.0025) for initial grading assumptions (per user note).
 - Codes: ASME B31.3, ACI 318, AISC, NFPA, NEC, OSHA, API 610, etc.
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3. Process mass balance and hydraulic flows (detailed)

(Full calculations retained from earlier report — summary numbers below.)

- Salt feed: **200,000 lb/hr**.
- Water required for 10% mass slurry: **1,800,000 lb/hr**.
- Total slurry mass: **2,000,000 lb/hr**.
- Slurry density: **8.9296 lb/gal** (10% NaCl @ 60 °F).
- Volumetric flow: **$\approx$ 223,975 gal/hr $\approx$ 3,733 GPM**.
- Pump hydraulic HP @ 100 ft TDH: **$\sim$ 100.9 HP**; motor HP (70% pump eff, 95% motor eff) $\approx$ **151.7 HP** $\rightarrow$ select **175 HP VFD**.

(See Appendix A for step-by-step math.)

4. Tank — geometry, internals, and utilities

- Volume = 10,000 gal = 1,336.81 ft³. H:D = 2:1 → **D ≈ 9.48 ft, H ≈ 18.95 ft**.
 - Internals: 4 baffles, top-entry agitator (15 HP), 36" manway, sample port, vent with flame arrestor if required, drain 3" valve. Insulation & heat trace to maintain 60 °F.
 - Loads (loaded): slurry mass = 89,296 lb; tank structure & internals estimated +10,000 lb → **total ≈ 99,296 lb** distributed to supports. Design supports per ACI & AISC. Foundation pad sizing prelim: 3 ft × 3 ft × 2 ft pads (4 pads).
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5. Belt conveyor — geometry, belt selection, sizing and drive

- 24 in belt (2 ft), length incl. allowances ≈ **101.5 ft** run, vertical rise ≈ **17.63 ft**, nominal belt speed 197 ft/min.
 - Salt volumetric feed ≈ **2,669.74 ft³/hr** ; loaded depth on 2 ft belt ≈ **1.355 in**.
 - Conveyor motor recommended **15 HP** (fixed speed). Include guards, pull cords, misalignment switches, E-stops, belts with abrasion cover, transfer skirt and chute lining.
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6. Discharge chute and transfer / dust control

- Lined skirted chute with impact plate and hood; transfer to tank top with sealed feed-in to minimize fugitive dust. Provision and space for small baghouse/local exhaust if dust control required (OPEN ITEM).
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7. Pump selection, hydraulic sizing, piping and NPSH considerations

- Design flow **3,733 GPM**, TDH **100 ft**, slurry SG **1.07**. Hydraulic HP ~100.86 HP. Motor selected **175 HP VFD** as planning size.
 - Pump: centrifugal slurry pump, 316 SS wetted parts; heavy-duty shaft and bearings. Consider lined casing or hardfacing if solids cause abrasion. Provide suction screening, straight suction run per vendor, and suction piping at least equal or larger (recommend **24 in suction**). Discharge piping **20 in** (confirm). Magnetic flowmeter for discharge. Compute NPSHa once final pump location and tank/pump elevations are available.
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8. Agitator selection and mixing basis

- Top-entry axial-flow / pitched blade agitator; power conservative selection **15 HP**. Baffles (4) to reduce vortexing. Shaft sealing and bearing per vendor; stuffing box or mechanical seal. Provide local access and lockout for maintenance.
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9. Instrumentation & controls (PLC architecture)

- Local PLC cabinet with HMI; analog and digital I/O for level, temp, flow, pressure, belt scale, E-stops, pull cords. Safety-rated I/O or separate hardwired safety circuits for E-stops as required. VFD for pump (175 HP), local starter or soft-start for conveyor & agitator. Instrument list retained (Appendix B).
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10. Structural & civil — platform, foundations, pedestals, site footprint & earthworks (UPDATED)

This section consolidates and finalizes the civil planning numbers, earthwork estimates, stormwater concept, grading, pavement, drainage, containment, survey and geotechnical budgets.

10.1 Site footprint & layout

- Working footprint estimate: $\sim 150 \text{ ft} \times 40 \text{ ft} = 6,000 \text{ ft}^2$ (includes conveyor, tank + platform, pump skid, maintenance clearances). Equipment footprint includes 20×20 ft area for tank + platform, conveyor run $\sim 120 \text{ ft}$ linear length with 6–8 ft width allowance for structure and walkway, pump skid 10×10 ft, plus access lane.

10.2 Foundation & pedestal design (prelim)

- Tank loaded weight $\approx 99,296 \text{ lb}$. Four pedestals $\rightarrow$ per-pedestal reaction $\approx 24,824 \text{ lb}$. With preliminary allowable bearing pressure **3,000 psf**, pad area $\approx 8.27 \text{ ft}^2$. We use **3 ft × 3 ft** pads (9 ft²) with 2 ft thickness as conservative preliminary pad; reinforced per ACI 318. Conveyor supports: 2×2×1.5 ft pads at $\sim 6\text{--}8 \text{ ft}$ spacing.

10.3 Earthwork & cut & fill (planning)

Detailed volumes (rounded):

- Tank pedestal pads (4): $72 \text{ ft}^3 = 2.67 \text{ cy}$

- Conveyor concrete supports (20): **120 ft³ = 4.44 cy**
- Pump foundation & anchor pockets: **200 ft³ = 7.41 cy**
- Platform slab (20×20×0.5 ft): **200 ft³ = 7.41 cy**
- Trench for main piping (120 ft × 3 ft × 4 ft): **1,440 ft³ = 53.33 cy**
- Tank excavation & working room: **1,000 ft³ = 37.04 cy**
- Miscellaneous: **500 ft³ = 18.52 cy**

Construction excavation/backfill total: $\approx 3,332 \text{ ft}^3 = 123.4 \text{ cy}$

General grading (pad leveling & swale shaping): $\sim 300 \text{ cy}$ moved (cut + fill operations).

Import/export contingency: $\sim 50 \text{ cy}$.

Planning total earthwork moved $\approx 473 \text{ cy}$ (net balance ~ 0 assumed; actual to be determined by topo).

Note: cut/fill balance will be refined from topo; the current numbers are planning-level estimates.

10.4 Grading & drainage concept

- Existing grade slope assumed **0.25%**. Site will be graded to provide positive drainage 2% away from platform to grassed swale. Perimeter grassed swale collects site runoff and transports to discharge point. Provide permanent erosion controls and sediment traps during construction.

10.5 Stormwater conceptual sizing (1-inch capture basis)

- Impervious area estimate: **4,800 ft²** (80% of 6,000 ft²).
- One-inch capture volume = $4,800 \text{ ft}^2 \times (1 \text{ in} = 1/12 \text{ ft}) = \mathbf{400 \text{ ft}^3 \approx 14.8 \text{ cy}}$ (minimum planning detention). Provide detention cell or equivalent infiltration/bioretenention with controlled outlet sized for local regulation. Recommend at least **400 ft³** storage on site; increase if local regs require.

10.6 Culvert and conveyance

- Driveway culvert: preliminary **12-inch RCP or HDPE** with headwalls & riprap at inlet/outlet. Provide inlet structure at low point; size per final rational method analysis.

10.7 Secondary containment & environmental controls

- Secondary containment sized to **110%** of tank volume: **10,000 gal × 110% = 11,000 gal = 1,466 ft³ $\approx 54.3 \text{ cy}$** . Provide coated concrete curb or lined containment basin (HDPE liner recommended because salt slurry is corrosive). Include sump/drain and recovery pump access. Provide spill response and containment signage.

10.8 Survey & geotechnical (required early items)

- Topographic & boundary survey (approx): **\$6,000** (planning).
 - Geotechnical borings (2 borings, lab tests, report): **\$6,000** (planning). These are essential and should be authorized immediately.
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11. Electrical — motor loads, transformer & distribution sizing

- Motor summary: Pump motor **175 HP**, conveyor **15 HP**, agitator **15 HP** → total motor HP = **205 HP** (~152.9 kW). With 30% auxiliary margin ≈ **198.8 kW** → apparent S ≈ **220.9 kVA** at 0.9 pf. Onsite transformer **480 V, 3φ, 500 kVA** adequate. Provide pump VFD (175 HP), starters for conveyors and agitator, MCC/main distribution, cables sized per NEC. Include arc flash study & coordination during detailed design.
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12. Thermal control — heat trace & insulation summary

- Maintain slurry components and piping at 60 °F. Use self-regulating heat trace on piping and selected tank shell tracing plus 1.5 in mineral wool insulation with aluminum cladding. Thermostat & overtemp protection required.
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13. Materials of construction, safety and access

- Tank: **304 SS**, process piping & wetted pump parts: **316 SS** (per earlier recommendations). Structural galvanized steel for platform etc. Walkways, stairs, handrails per OSHA. Lockout-tagout, confined-space signage, emergency shutdowns integrated in PLC.
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14. Open items (data required for final design)

(Repeated and expanded; these must be resolved to finalize design and costing)

1. Final confirmation of feed rate and slurry concentration (100 st/hr; 10% mass confirmed).
2. Final pump installation elevation & suction geometry (critical for NPSHa).
3. Site coordinates & detailed topo (required to finalize cut/fill and stormwater).
4. Local stormwater design standards (Charlotte/Mecklenburg) — design storms, IDF curves, allowable post-construction discharge.
5. Confirm dust control requirement at conveyor discharge (baghouse vs local hood).
6. Confirm municipal fire flow (decide whether fire pump required).
7. Final pipe material decision for large-diameter piping (20–24 in 316 SS vs alternative).
8. Confirm if redundancy is required for pump (1 vs 2).
9. Confirm power tie-in location & transformer availability.
10. Confirm tank venting pressure limits and whether tank will operate pressurized or atmospheric.

15. Deliverables & schedule summary (final)

Deliverables

- Updated full Design Criteria Report (this document).
- Detailed calculations & spreadsheets (including civil cut/fill and hydrology).
- PFD and P&ID sketches.
- Equipment datasheets & vendor selection templates.
- Structural drawings for platform & pedestals.
- Electrical one-line, MCC layout.
- Instrument index & PLC I/O list.
- Bill of Materials (BOM) & procurement packages.
- Installation estimate and commissioning plan.

Preliminary schedule (indicative)

- Kickoff & data confirmation: 2 weeks (survey & geotech authorized concurrently).
- Detailed design & vendor selection: 8–10 weeks.
- Procurement & fabrication: 10–14 weeks.
- Site civil & foundations: 4–6 weeks.
- Mechanical installation: 4–6 weeks.
- Commissioning & performance testing: 2 weeks.

16. Preliminary cost estimate (itemized, with 30% contingency) — UPDATED

All numbers are planning-level, US\$ 2025. The table below aggregates the original equipment block numbers and the expanded civil & safety/utility items added in this final report.

16.1 Itemized pre-contingency estimates

Category	Estimate (USD) — Pre-contingency
Mechanical equipment (conveyor, chute, belt scale, small items)	\$45,000
10,000 gal 304 SS tank + agitator + internals	\$70,000
Pump (centrifugal stainless slurry pump), skid & VFD (175 HP)	\$130,000
Process piping (316 SS, 20–24 in) — suction/discharge, valves, fittings, insulation, heat trace	\$210,000
Structural steel (platform, stair tower, ladders, grating, galvanizing)	\$60,000
Civil (survey, geotech, earthwork, pavement, SWM, culvert, erosion control)	\$82,600

Category	Estimate (USD) — Pre-contingency
Concrete foundations & civil pad items (pads, pedestals, slab)	\$10,000
Instrumentation & controls (PLC, HMI, VFD, flowmeter, level, temp, pressure, valves, wiring)	\$75,000
Electrical (MCC, starters, motor cables, distribution tie-in)	\$45,000
Mechanical installation & field labor (mechanical, piping, welding, electrical)	\$140,000
Safety & containment (secondary containment 110%, fire protection, spill control)	\$94,500
Instrument air compressor & utilities (installed)	\$11,000
Indirects (engineering, permitting, commissioning, vendors)	\$60,000
Subtotal (pre-contingency)	\$1,033,100

Note: the subtotal here is slightly different from earlier versions because civil and safety blocks were expanded and the optional fire pump is included in safety. If municipal fire flow removes the need for on-site fire pump, subtract approx **\$50,000** from safety and subtotal.

16.2 Contingency and grand total

- **30% contingency:** $\$1,033,100 \times 0.30 = \$309,930$
- **Grand total (with 30% contingency):** **\$1,343,030**

This grand total differs from the earlier \$1,147,380 because the updated cost table includes a more thorough breakout (the civil block here includes more items and the safety block includes an optional fire pump). To reconcile to earlier totals, you can remove optional items (fire pump) and/or adjust piping material choices. Below we provide an alternative grand total excluding optional fire pump and using the more conservative prior subtotal to match earlier reported totals.

16.3 Alternative planning total (match earlier presentation, optional fire pump excluded)

- Subtotal (pre-contingency, excluding on-site fire pump): **\$882,600** (approx earlier subtotal)
- 30% contingency: **\$264,780**
- **Total (with 30% contingency):** **\$1,147,380** (this matches the earlier consolidated number and is appropriate if municipal fire flow is available and piping choices remain as earlier).

Recommendation: Use the \$1.15M figure as base planning budget if you anticipate municipal fire flow and confirm piping materials as 316 SS for process piping. If you must include a dedicated fire pump, use the \$1.34M figure.

17. Codes, standards & references

- ASME B31.3 — Process Piping

- ASME Section VIII — Pressure vessels (if applicable)
 - API pump standards / API 610 where applicable
 - AISC Steel Construction Manual
 - ACI 318 — Reinforced concrete design
 - AWS D1.1 — Structural welding code
 - ANSI/ASME B16.5 — Flanges & flanged fittings
 - NFPA 70 (NEC) — Electrical code
 - OSHA 1910 — Safety and guarding
 - NACE & ASTM material standards
 - Local Charlotte/Mecklenburg stormwater and permitting requirements (to be used in detailed design)
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Appendices (summaries, calculations & tables)

Appendix A — Key calculation highlights

- Salt feed: **100 st/hr = 200,000 lb/hr**
- Water required for 10% mass slurry: **1,800,000 lb/hr**
- Total slurry mass: **2,000,000 lb/hr**
- Slurry density: **8.9296 lb/gal**
- Slurry volumetric flow: **223,975 gal/hr $\approx$ 3,733 GPM**
- Pump hydraulic HP @ 100 ft TDH: **$\sim$ 100.9 HP**
- Recommended pump motor (prelim): **175 HP VFD**
- Tank geometry: $D \approx 9.48$ ft; $H \approx 18.95$ ft (for 10,000 gal, $H:D = 2:1$)
- Conveyor volumetric feed: **2,669.74 ft³/hr**, belt loaded depth $\approx$ **1.355 in** on 24 in belt.

Appendix B — Preliminary instrument list

(Selected)

- Tank level transmitter (4–20 mA), high-high level switch, RTD temperature transmitter.
- Discharge flowmeter (magnetic for slurry), suction & discharge pressure transmitters.
- Belt scale (tons/hr), speed sensor, belt misalignment switch, pull cords, E-stops.
- PLC + HMI, VFD for pump, starters for conveyor & agitator, field junction boxes, cable trays.

Appendix C — Civil tables & earthwork summary (planning)

- Construction excavation/backfill: $\approx$ **123.4 cy**
- General grading moved: $\sim$ **300 cy**
- Import/export contingency: $\sim$ **50 cy**
- **Total earthwork moved: $\sim$ 473 cy**

Appendix D — Containment sizing

- Tank volume 10,000 gal → 110% containment = **11,000 gal** → **1,466 ft³ ≈ 54.3 cy**. Provide a lined and coated containment area with sump and recovery pump access.

Appendix E — Costs summary (condensed)

- Grand total (option A — municipal fire flow, no on-site fire pump): **\$1,147,380** (with 30% contingency).
 - Grand total (option B — including on-site fire pump): **\$1,343,030** (with 30% contingency).
 - Key drivers: **316 SS large-diameter piping**, pump & VFD, mechanical installation labor, and stormwater/containment provisions.
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Final remarks & next actions (urgent)

1. **Authorize survey & geotechnical immediately** (budget ≈ \$12k combined). These are gating items for civil, foundation & NPSHa verification.
 2. **Confirm municipal fire flow** (contact local fire marshal) to determine whether on-site fire pump is required (a \$50k planning allowance exists in safety block).
 3. **Confirm pump location elevation** relative to tank inlet to compute NPSHa.
 4. **Decide piping material strategy** (316 SS vs lined carbon steel) — piping is the major cost driver and will materially affect budget.
 5. If you want, I will produce (no extra data required):
 - o A printable **one-page civil plan & cut/fill table** for permitting submittal (simple sketch), or
 - o A refined **cost estimate** where I remove optional items and produce an equipment procurement list by vendor class.
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If you'd like, I'll immediately generate the **one-page civil plan sketch with a cut/fill table and the final cost table formatted to a PDF** for submittal and budgeting. I can also produce a printable single-page executive summary for leadership. Which deliverable would you like next?